Project Proposal – Extracting Action Items from Unstructured & Noisy Meeting Data

Large Language Models and its Applications

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Motivating Use Case

The Problem

- Meetings are critical, but tracking the **missions** generated within them is manual, slow, and error-prone
- A „Productivity Black Hole“

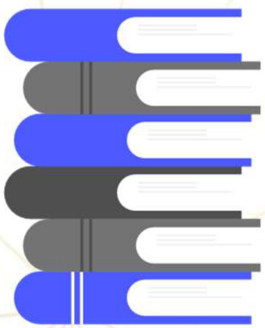
The Challenge

- Unstructured Data
- Current Solution

Solution (The Project)

- **Our Solution**: Automated Action Item Extraction System specifically designed for Noisy Environments.
- **The Goal:** Not just identifying tasks, but structuring them (Who, What, When) and flagging uncertainty.

Project Task Description



Task Definition

- **ML Task:** Relational Extraction & Uncertainty Estimation (Not just Classification).

- **Input**: A context-rich meeting segment containing "organizational noise" (slang, interruptions, code-switching).

- **Output:** Structured JSON object {Assignee, Action, Deadline} + Confidence Score

- **Goal:** Extract actionable items while filtering out "hallucinations" or ambiguous commitments.

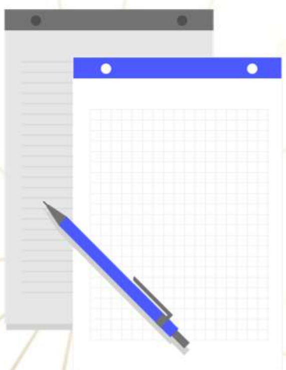
Models and Methods

- **Step 1:** Data Generation (LLM simulator for noisy meetings).
- **Step 2:** Extraction Model (Fine-tuned Llama/Mistral).
- **Step 3:** Validation Layer (Checking for missing info/hallucinations)

Data Specification

- **The Gap:** No dataset exists for Israeli corporate culture.
- **Our Data:** Generating 500+ examples of "Messy" meetings.
- **Attributes:** Slang, Code-Switching (Hebrew/English), Interruptions.

Evaluation Metrics & KPIs



- **Extraction Accuracy (Exact Match):** Measuring the percentage of correctly extracted JSON fields (Correct Assignee + Correct Action Description).
- **Hallucination Rate (Reliability):** Measuring how often the model invents a task where none exists (False Positives) – crucial for user trust.
- **Confidence Calibration:** Validating that "Low Confidence" scores actually correlate with vague or ambiguous inputs.
- **Baseline:** Comparing our Fine-Tuned model against standard Zero-Shot GPT-4 performance.